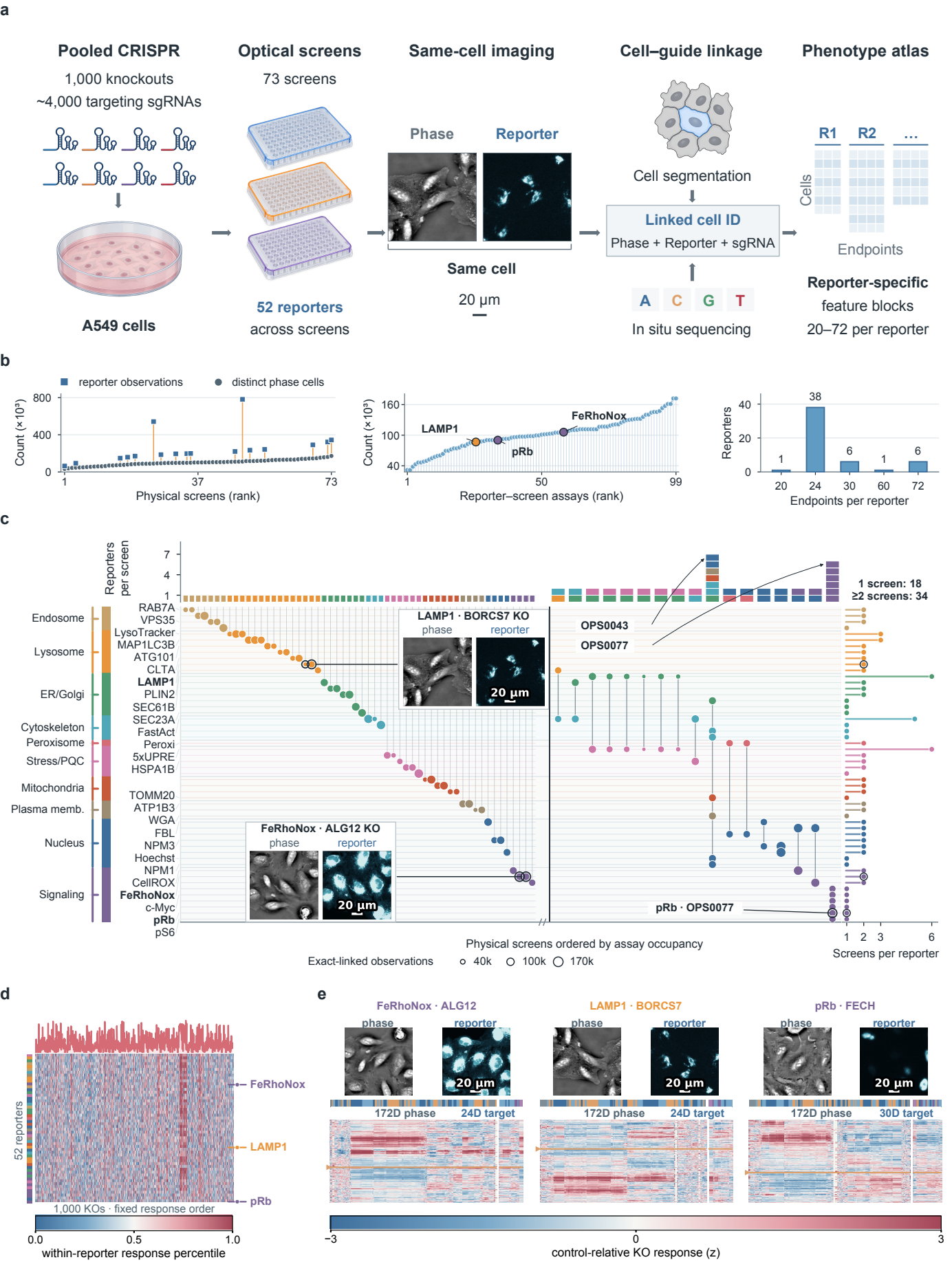

a
Pooled CRISPR
1,000 knockouts
~4,000 targeting sgRNAs
A549 cells
Optical screens
73 screens
52 reporters
across screens
Same-cell imaging
Phase
Reporter
Same cell
20 µm
Cell–guide linkage
Cell segmentation
Linked cell ID
Phase + Reporter + sgRNA
A C G T
In situ sequencing
Phenotype atlas
R1 R2 ...
Cells
Endpoints
Reporter-specific
feature blocks
20–72 per reporter
b
reporter observations
distinct phase cells
Count (×10³)
Physical screens (rank)
LAMP1
pRb
FeRhoNox
Reporter–screen assays (rank)
Reporters
Endpoints per reporter
c
Reporters per screen
Endosome
Lysosome
ER/Golgi
Cytoskeleton
Peroxisome
Stress/PQC
Mitochondria
Plasma memb.
Nucleus
Signaling
RAB7A
VPS35
LysoTracker
MAP1LC3B
ATG101
CLTA
LAMP1
PLIN2
SEC61B
SEC23A
FastAct
Peroxi
5xUPRE
HSPA1B
TOMM20
ATP1B3
WGA
FBL
NPM3
Hoechst
NPM1
CellROX
FeRhoNox
c-Myc
pRb
pS6
LAMP1 · BORCS7 KO
phase
reporter
20 µm
FeRhoNox · ALG12 KO
OPS0043
OPS0077
1 screen: 18
≥2 screens: 34
pRb · OPS0077
Physical screens ordered by assay occupancy
Exact-linked observations
40k
100k
170k
Screens per reporter
d
52 reporters
FeRhoNox
LAMP1
pRb
1,000 KOs · fixed response order
within-reporter response percentile
e
FeRhoNox · ALG12
LAMP1 · BORCS7
pRb · FECH
phase
reporter
20 µm
172D phase
24D target
30D target
control-relative KO response (z)